Engine Model(s): PT6A

73-10-08

PNEUMATIC LINE (Py) - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00, STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, 0.025 inch diameter, and will not be called out in instructions.

2. Consumable Materials

Not Applicable

3. Special Tools

Not Applicable

4. Fixtures, Equipment and Supplier Tools

Not Applicable

5. Removal/Installation

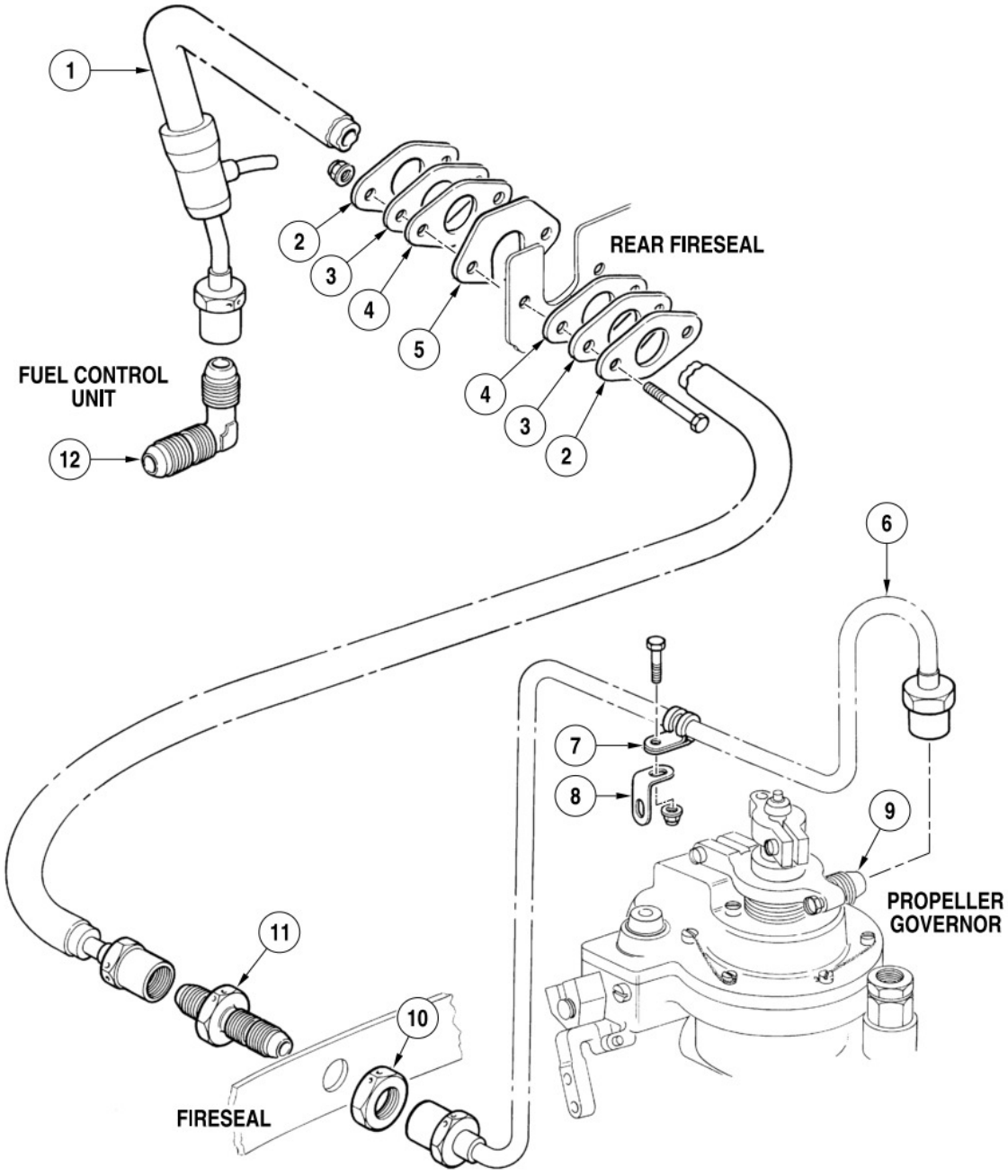
CAUTION: HOLD THE FITTING CORRECTLY IN POSITION WHEN YOU TORQUE THE COUPLING NUT.

NOTE: It is important to make sure that all solid P3 lines are properly and correctly fitted before the coupling nuts are tightened. The coupling nuts should be seated without the use of tools (finger tight) before torquing. The installation of any pipe brackets should not distort the pipe in any way.

A. Removal (Ref. Fig. 201)

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Figure 201 Removal/Installation of Py Air Pressure Lines



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1. Pneumatic Heated Rear Line
2. Retaining Plate
3. Insulation
4. Seal
5. Slotted Retaining Plate
6. Pneumatic Front Line
7. Clamp
8. Bracket
9. Straight Nipple
10. Locknut
11. Bulkhead Coupling
12. Elbow (Ref.)

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(1) Detach the electrical cable of the heated rear line (1) and disconnect the leads from the connector at Flange G (Ref. 73-10-07).

(2) Remove the two self-locking nuts and bolts, and detach the sealing components (2), (3) and (4) from the rear fireseal. Remove the slotted retaining plate (5).

(3) Disconnect the heated line (1) from the elbow (12) on the FCU and bulkhead coupling (11) at the center fireseal.

(4) Remove the heated line from the slot in the rear fireseal and remove the sealing components from the line.

(5) Remove the self-locking nut and bolt securing the clamp (7) on the front line (6) to the bracket (8).

(6) Disconnect the front line (6) from the straight nipple (9) on the propeller governor and the bulkhead coupling (11) at the center fireseal and remove the line. Remove the clamp (7) from the line.

(7) If on visual inspection, the bulkhead coupling (11) appears to be damaged, remove the coupling from the center fireseal by removing the locknut (10).

B. Installation of Pneumatic Line (Ref. Fig. 201)

CAUTION: HOLD THE FITTING CORRECTLY IN POSITION WHEN YOU TORQUE THE COUPLING NUT.

NOTE: It is important to make sure that all solid P3 lines are properly and correctly fitted before the coupling nuts are tightened. The coupling nuts should be seated without the use of tools (finger tight) before torquing. The installation of any pipe brackets should not distort the pipe in any way.

(1) If the bulkhead coupling (11) was removed, reinstall the coupling on the center fireseal and secure it with the locknut (10). Tighten the locknut 38 to 42 lb.in.

NOTE: Do not lockwire connections of pneumatic lines; lockwiring must be accomplished after the system pressure check.

(2) Install the front line (6) by connecting the line to the straight nipple (9) on the propeller governor and the bulkhead coupling (11). Tighten the coupling nuts 90 to 100 lb.in.

(3) Secure the line to the bracket (8) with a clamp (7), bolt and self-locking nut. Tighten the nut 36 to 40 lb.in.

(4) Sequentially install the seals (4), insulation (3) and retaining plates (2) on the heated rear line (1).

(5) Install the line (1) by locating the line in the slot of the rear fireseal with sealing components disposed as shown, on each side of the fireseal; connect the line to the elbow (12) on the FCU and the bulkhead coupling (11) at the center fireseal. Tighten the coupling nuts 90 to 100 lb.in.

(6) Install the slotted retaining plate (5) on the line on the rear face of the rear fireseal. Attach the slotted plate and sealing components with two bolts and self-locking nuts. Tighten the

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nuts until firm contact is obtained, plus 180 degrees.

(7) Connect the leads of the electrical cable of the heated line (1) to the connector at Flange G, and secure the cable (Ref. 73-10-07).

6. Cleaning/Painting

A. Cleaning of Air Tube(s)

(1) Clean air tube(s) (Ref. 73-10-07, MAINTENANCE PRACTICES, Cleaning/Painting).

7. Inspection/Check

A. Inspection

(1) Insulated Tube Assemblies

(a) Examine tube assemblies (Ref. 70-00-00, STANDARD PRACTICES - INSPECTION).

(b) Blend out minor damage that does not exceed specified limits.

(c) Replace the tube assemblies damaged beyond specified limits.

(d) Inspect heated rear pneumatic line (Ref. 73-10-07, Inspection/Check).

(e) Examine insulation on pneumatic tube assemblies for cuts in the outer rubber sheath. Cuts up to three inches long may be repaired (Ref. 73-10-17, Approved Repairs).

(f) If surface damage exceeds specified limits above, but depth is superficial, sheathing may be repaired by replacement of the damaged portion (Ref. 73-10-07, Approved Repairs).

8. Approved Repairs

A. General Repairs

(1) Repair damaged insulation on the heated rear line (Ref. 73-10-07).

(2) Damaged threads of the bulkhead coupling may be repaired by chasing. Raised metal on the hexagon flats may be removed by light stoning.

(3) Damaged electrical connector may be replaced (Ref. 73-10-07).

9. Adjustment/Test

Do a pressure test of the pneumatic system (Ref. 73-10-07).

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